

A Lightweight ID-less Authentication Mechanism for IoT-UAV Communication

Project Title

A Lightweight ID-less Authentication Mechanism for IoT-UAV Communication Using Timestamp Division and HMAC

Specifications

- 1 Simulation Tool: ns-3
- 2 Programming Language: C++
- 3 Network Standard: IEEE 802.11b (Ad hoc mode)
- 4 Communication Protocol: UDP
- 5 Number of IoT Devices: Configurable (1–50)
- 6 UAV Nodes: 1 (Edge/MEC node)
- 7 Packet Size: ~190 bytes
- 8 Authentication Delay: ~13 ms
- 9 Simulation Time: 10 seconds

Introduction

The rapid growth of IoT devices and UAV-based edge computing introduces significant security challenges in wireless communication. Traditional authentication methods either expose device identity or require heavy computation. This project proposes a lightweight ID-less authentication mechanism using SHA-256, timestamp division, and HMAC-SHA256 to ensure privacy, security, and low-latency communication.

Implementation

- 1 Multiple IoT devices are created as nodes in ns-3
- 2 A UAV node acts as the authentication verifier
- 3 UDP communication over IEEE 802.11b is used
- 4 Authentication delay (~13 ms) is simulated
- 5 Lightweight packet size (~190 bytes) is maintained
- 6 Network delay and performance are evaluated

Steps to Run

- 1 Place the file in ns-3/scratch/iot-uav-auth.cc
- 2 Run: `./waf build`
- 3 Run: `./waf --run scratch/iot-uav-auth`
- 4 Optional: `./waf --run "scratch/iot-uav-auth --nDevices=20"`

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